

Temperature Dependent Properties

$$T_m = \frac{T_{m,i} + T_{m,o}}{2}$$

$$\frac{Nu}{Nu_{lm}} = f\left(\left(\frac{u_{ax}}{u_{lm}}\right)^{n_1}\right) = g\left(\left(\frac{T_w}{T_m}\right)^{n_2}\right)$$

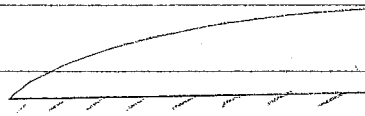
In a typical design problem, we may not know $T_{m,o}$. It is a strong function of the amount of heat transfer.

Laminar Boundary Layers

Concentration

W_0

u
 T_0 →



$$u_w = 0$$

$$T_w \neq T_0$$

$$W_w = W_0$$

Viscous effects are very small outside BL.
Outside flow is inviscid. Inviscid fluid will slip on a boundary.

ex u_∞



flow is symmetric for inviscid fluid.

$$u_w|_{r=a} = u_{slip}(\theta)$$

Temperature Dependent Properties

$$T_m = \frac{T_{m,i,n} + T_{m,o}}{2}$$

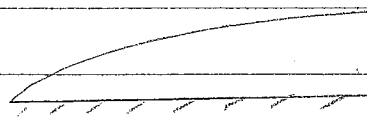
$$\frac{Nu}{Nu_{lm}} = f\left(\left(\frac{\mu_{lm}}{\mu_s}\right)^{n_1}\right) = g\left(\left(\frac{T_w}{T_m}\right)^{n_2}\right)$$

In a typical design problem, we may not know $T_{m,o}$. It is a strong function of the amount of heat transfer.

Laminar Boundary Layers

Concentration W_0

u
 T_0 →



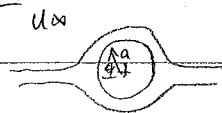
$$u_w = 0$$

$$T_w \neq T_0$$

$$W_w = W_0$$

Viscous effects are very small outside BL.
Outside flow is inviscid. Inviscid fluid will slip on a boundary.

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flow is symmetric for inviscid fluid.

$$u_\infty|_{r=a} = u_{slip}(\theta)$$

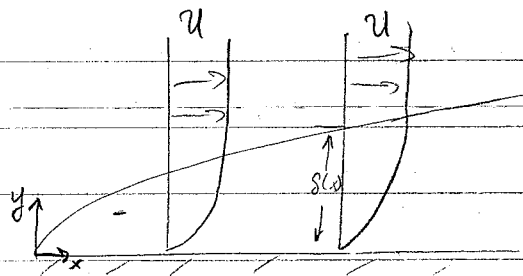
Solution breaks into two parts:

1) External flow which is inviscid. It doesn't know a fixed boundary is present.

2) Boundary Layer Solution. with no-slip at plate.

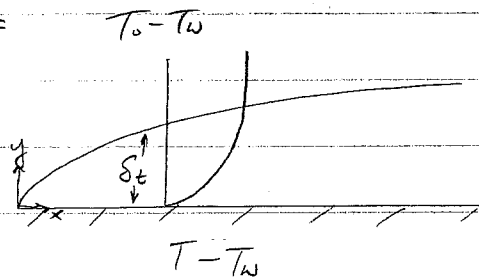
velocity BL

u
 $T_\infty \rightarrow$
 $u_\infty \rightarrow$



Temperature BL

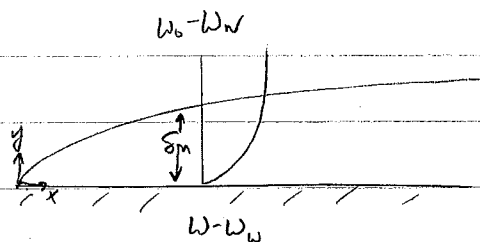
u
 $T_\infty \rightarrow$
 w_∞



$$Pe = \frac{uL}{\alpha}$$

Concentration BL

u
 $T_\infty \rightarrow$
 w_∞



The NS equations are

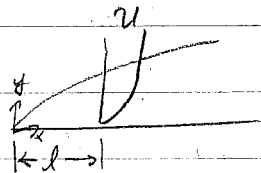
$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$$

$$\rho \left(u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} \right) = - \frac{\partial p}{\partial x} + \mu \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right) + B_x$$

$$\rho \left(u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} \right) = - \frac{\partial p}{\partial y} + \mu \left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} \right) + B_y$$

At $y=0$, $u=0$
 $v=0$

At $y \rightarrow \infty$, $u \rightarrow U$
 $v \rightarrow 0$



Nondim scales:

$$x' = \frac{x}{l}$$

$$u' = \frac{u}{U}$$

$$P' = \frac{P}{\rho U^2}$$

$$Re = \frac{U l}{\nu}$$

$$y' = \frac{y}{l}$$

$$v' = \frac{v}{U}$$

$$B' = \frac{B}{\rho U^2 / l}$$

The nondimensionalized equations are

$$\frac{\partial u'}{\partial x'} + \frac{\partial v'}{\partial y'} = 0$$

$$u' \frac{\partial u'}{\partial x'} + v' \frac{\partial u'}{\partial y'} = B'_x - \frac{\partial P'}{\partial x'} + \frac{1}{Re} \left(\frac{\partial^2 u'}{\partial x'^2} + \frac{\partial^2 u'}{\partial y'^2} \right)$$

$$u' \frac{\partial v'}{\partial x'} + v' \frac{\partial v'}{\partial y'} = B_y' - \frac{\partial P'}{\partial y'} + \frac{1}{Re} \left(\frac{\partial^2 v'}{\partial x'^2} + \frac{\partial^2 v'}{\partial y'^2} \right)$$

$$\text{At } y'=0, u'=0 \\ v'=0$$

$$\text{At } y' \rightarrow \infty, u' \rightarrow 1 \\ v' \rightarrow 0$$

Since y' is of the order of δ' ,

$$y' = O(\delta') \quad , \quad \delta' = \frac{\delta}{l}$$

$$\text{At BL, } \delta' \rightarrow 0$$

$\Rightarrow$ Any term with y' in the denominator will be very large, i.e. $\frac{\partial v'}{\partial y'}$

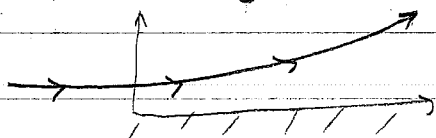
Unless numerator is also small!!

$$\left. \begin{array}{l} u' = O(1) \\ v' = O(1) \end{array} \right\} \text{Look at continuity}$$

$$\frac{\partial u'}{\partial x'} + \frac{\partial v'}{\partial y'} = 0$$

$$\frac{O(1)}{O(1)} + \frac{\partial v'}{O(\delta')} = 0$$

This means that $v' = O(\delta')$ in order to satisfy continuity.



Look at X-momentum,

$$u' \frac{\partial u'}{\partial x'} + v' \frac{\partial u'}{\partial y'} = B_v' - \frac{\partial p'}{\partial x'} + \frac{1}{Re} \left(\frac{\partial^2 u'}{\partial x'^2} + \frac{\partial^2 u'}{\partial y'^2} \right)$$

We get $\delta' = O(Re^{-\frac{1}{2}})$ and

Look at y -moments,

$$u' \frac{\partial v'}{\partial x'} + v' \frac{\partial v'}{\partial y'} = B_y' - \frac{\partial p'}{\partial y'} + \frac{1}{S'^2} \left(\frac{\partial^2 v'}{\partial x'^2} + \frac{\partial^2 v'}{\partial y'^2} \right)$$

Removing terms which are negligible ($O(\delta')$ makes it small),
NS eqns for BL become

$$\frac{\partial u'}{\partial x'} + \frac{\partial v'}{\partial y'} = 0$$

$$u' \frac{\partial u'}{\partial x'} + v' \frac{\partial u'}{\partial y'} = B'_x - \frac{\partial p'}{\partial x'} + \frac{1}{Re} \frac{\partial^2 u'}{\partial y'^2}$$

$$0 = B_y' - \frac{\partial \mathcal{L}'}{\partial y'} \quad \text{or order } \delta'$$

$$At y' = 0, u' = 0$$

At $y' \rightarrow \infty$, $\psi' \rightarrow 2$

Outside BL,

$$u \frac{\partial u}{\partial x} = B_x - \frac{\partial p}{\partial x} \quad \checkmark \text{ for flat plate}$$

Which is the streamwise derivative of the Bernoulli's eqn.

There is no change of pressure across BL + $\frac{\partial p}{\partial y} = 0$
change in x is driven by external flow.

The temperature diffusion (with α) and concentration diffusion (with D) have very similar solutions. We will just look at Temperature solution.

$$\rho C_p (u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y}) = k (\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2}) + \Phi + \beta T (u \frac{\partial p}{\partial x} + v \frac{\partial p}{\partial y})$$

$$\text{At } y=0, T=T_w$$

$$\text{At } y \rightarrow \infty, T=T_\infty$$

$$\theta = \frac{T - T_\infty}{T_w - T_\infty}$$

The nondim form is

$$u' \frac{\partial \theta}{\partial x'} + v' \frac{\partial \theta}{\partial y'} = \frac{1}{Re Pr} \left(\frac{\partial^2 \theta}{\partial x'^2} + \frac{\partial^2 \theta}{\partial y'^2} \right) + \frac{E}{Re} \Phi' + \beta T' E (u' \frac{\partial p'}{\partial x'} + v' \frac{\partial p'}{\partial y'})$$

ORDERS:

$O(1)$

$\frac{\delta'}{\delta_T}$

$\frac{\delta'^2}{r}$

$(1 + \frac{1}{\delta'^2})$

largest term

$O(1)$

$(1 + \delta' \delta')$

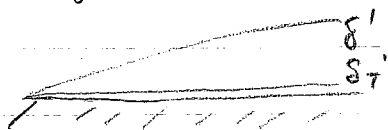
Look at 1st term on RHS;

$$\frac{\delta'^2}{Pr \delta_T'^2} = O(1)$$

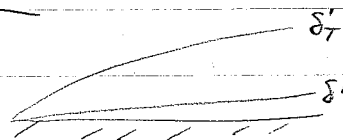
$$\boxed{\frac{\delta_T'}{\delta'} \sim Pr^{\frac{1}{2}}}$$

If $Pr \approx 1$, both δ' & δ_T' are equal $\Rightarrow$ BL thicknesses are the same.

Pr large



$Pr \rightarrow 0$



After sorting out orders, inside BL:

$$\rho \left(\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} \right) = B_x - \frac{\partial p}{\partial x} + \frac{\partial}{\partial y} \left(\mu \frac{\partial u}{\partial y} \right)$$

$$0 = B_y - \frac{\partial p}{\partial y} \rightarrow \text{pressure is hydrostatic.}$$

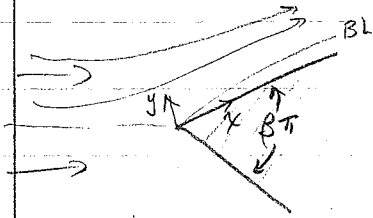
$$\rho C_p \left(\frac{\partial T}{\partial t} + u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y} \right) = \frac{\partial}{\partial y} \left(k \frac{\partial T}{\partial y} \right) + u \left(\frac{\partial u}{\partial y} \right)^2 + B_T \left[\frac{\partial p}{\partial t} + u \frac{\partial p}{\partial x} \right]$$

$$\rho \left(\frac{\partial w}{\partial t} + u \frac{\partial w}{\partial x} + v \frac{\partial w}{\partial y} \right) = \frac{\partial}{\partial y} \left(\rho D \frac{\partial w}{\partial y} \right)$$

Outside BL:

$$\rho \left(\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} \right) = B_x - \frac{\partial p}{\partial x}$$

Flow with Pressure Gradients (Wedge Flow)



$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$$

$$u u_x + v u_y = -\frac{1}{\rho} p_x + \nu u_{yy}$$

$$u T_x + v T_y = \alpha T_{yy} + \frac{\nu}{C_p} u_{yy}^2$$

Look at Bernoulli's Egn:

$$\frac{p}{\rho} + \frac{u^2}{2} = \text{const}$$

$$\frac{1}{\rho} \frac{\partial p}{\partial x} + u \frac{\partial u}{\partial x} = 0$$

First step is to solve for external inviscid flow:

$$\boxed{u = c x^m}, \quad m = \frac{\beta}{2-\beta}, \quad \beta = \frac{2m}{1+m}$$

obtained by solving Euler Equations:

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$$

$$u u_x + v u_y = -\frac{1}{\rho} p_x$$

$$u v_x + v v_y = -\frac{1}{\rho} p_y$$

$$p + \frac{u^2}{2} = \text{const}$$

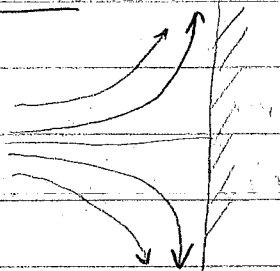
$$u = \psi_y \quad v = -\psi_x$$

$$u = \phi_x \quad v = \phi_y$$

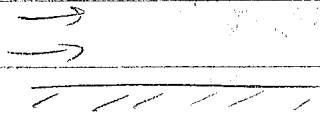
$$\Rightarrow \nabla^2 \phi = 0$$

$$\nabla^2 \psi = 0$$

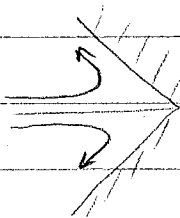
for $\beta = 1$



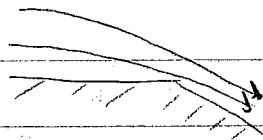
for $\beta = 0$



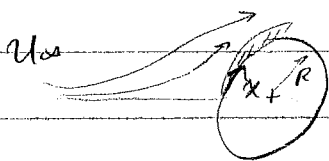
for $\beta = 3/2$



for $\beta < 0$



Flow over cylinder/sphere



2D: $U \sim 2U_\infty \frac{x}{R}$ at nose

3D: $U \sim \frac{3}{2} U_\infty \frac{x}{R}$

Given the potential flow field

$$u = cx^m$$

($m=0 \Rightarrow$ Blasius)

given C & m , we can solve

$$-\frac{1}{\rho} \frac{dp}{dx} = u \frac{du}{dx} = u cm x^{m-1} = \frac{m}{x} u^2$$

The BL equations are

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$$

$$u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} = -\frac{1}{\rho} \frac{dp}{dx} + \nu \frac{\partial^2 u}{\partial y^2}$$

$$u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y} = \alpha \frac{\partial^2 T}{\partial y^2} + \frac{\nu}{c_p} \left(\frac{\partial u}{\partial y} \right)^2$$

$$y=0 : u=v=0, \text{ either } T=T_w \text{ or } \frac{\partial T}{\partial y}=0, T_{\text{ins}}=?$$

$$y \rightarrow \infty : u \rightarrow U, T \rightarrow T_{\infty}$$

The similarity transformation is

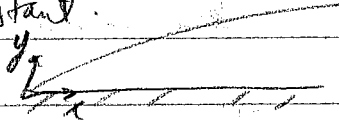
$$\eta = y \sqrt{\frac{U}{\nu x}}$$

$$\psi = \sqrt{\nu U x} F(\eta)$$

FLAT PLATE

Let $\frac{\partial p}{\partial x} = 0$ since Unconstant.

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0.$$



$$u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} = \nu \frac{\partial^2 u}{\partial y^2}$$

$$\text{At } y=0, \quad u=0 \\ v=0$$

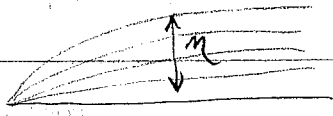
$$\text{At } y \rightarrow \infty, \quad u=U$$

BL solution lost $\frac{\partial^2}{\partial x^2} \Rightarrow$ flow is parabolic in x .

Look for similarity variable

$$\eta(x, y) = \text{constant.}$$

ex $\eta = \frac{y}{\sqrt{x}} \Rightarrow y = \text{constant} \sqrt{x}$



$$\frac{u}{U} = f(\eta) \text{ only!!}$$

This transformation allows $2D (x, y) \rightarrow 1D (\eta)$

See Appendix D for Similarity Transform

We can transform using Streamfunction

$$u = \frac{\partial \psi}{\partial y}, \quad v = -\frac{\partial \psi}{\partial x}$$

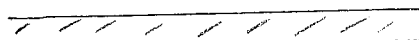
Insert into x-momentum

$$\psi_y \psi_{yx} - \psi_x \psi_{yy} = \nu \psi_{yyy}$$

$$y=0: \psi_x = \psi_y = 0, \quad y \rightarrow \infty: \psi_y \rightarrow u$$

$u \rightarrow$

ψ constant along plate
 $\Rightarrow$ plate is a streamline



We choose arbitrary value ψ at plate.

$\therefore$ We choose $\psi = 0$ at plate $\Rightarrow \psi_x = 0$.

We search for (Outline of Appendix D)

$$(x, y) \rightarrow (x, \eta) \quad \text{where } \eta(x, y)$$

$$\psi = H(x) F(\eta)$$

$$\eta = y g(x)$$

$$\frac{\partial}{\partial x} \Big|_{\eta} \rightarrow \frac{\partial}{\partial x} + \frac{\partial \eta}{\partial x} \Big|_y \frac{\partial}{\partial \eta}$$

$$\frac{\partial \eta}{\partial x} \Big|_y = y g'$$

$$\frac{\partial}{\partial y} \Big|_{\eta} \rightarrow \frac{\partial}{\partial \eta} \frac{\partial \eta}{\partial y} \Big|_x$$

$$\frac{\partial \eta}{\partial y} \Big|_x = g$$

Take everything & insert into Stream function equation. Then solve for $F(\eta)$ to find similarity transformation.

A different approach to find η Scales:

$$x_0, y_0, \psi_0$$

$$x' = \frac{x}{x_0}, \quad y' = \frac{y}{y_0}, \quad \psi' = \frac{\psi}{\psi_0}$$

Insert into ψ eqn.

$$\frac{\psi_0^2}{y_0^2 x_0} (\psi'_{y'} \psi'_{y'x'} - \psi'_{x'} \psi'_{y'y'}) = \frac{2\tau_0}{y_0^3} \psi'_{y'y'y'}$$

$$(\psi'_{y'} \psi'_{y'x'} - \psi'_{x'} \psi'_{y'y'}) = \frac{\nu x_0}{y_0 \psi_0} \psi'_{y'y'y'}$$

$$y'=0: \psi' = \psi'_{y'} = 0 \quad y' \rightarrow \infty: \psi' \sim \frac{u y_0}{\psi_0}$$

Let

$$\psi' = f_1 \left(x', y', \frac{y_0 \psi_0}{\nu x_0}, \frac{u y_0}{\psi_0} \right)$$

$$\frac{\psi'}{\psi_0} = f_1 \left(\frac{x}{x_0}, \frac{y}{y_0}, \frac{y_0 \psi_0}{\nu x_0}, \frac{u y_0}{\psi_0} \right)$$

$$= f_2 \left(\frac{x}{x_0}, \frac{y}{y_0}, \frac{y_0 \psi_0}{\nu x_0}, \frac{u y_0}{\psi_0} \right)$$

° independent of x_0

$$\frac{\psi}{\psi_0} = f_3 \left(\frac{y}{y_0}, \frac{y \psi_0}{\nu x}, \frac{u y}{\psi_0} \right) \quad (\text{indep. of } y_0)$$

$$\frac{\psi}{u y} = f_4 \left(\frac{y^2 u}{\nu x}, \frac{u y}{\psi_0} \right) \quad (\text{No } \psi_0 \text{ dependence})$$

$$\frac{\psi}{u y} = f_5 \left(\frac{u y^2}{\nu x} \right)$$

We take $\boxed{\eta = y \sqrt{\frac{u}{\nu x}}}$

Then ψ becomes

$$\psi = u y f_6(\eta)$$

We want the form $f_6(\eta) = \frac{F(\eta)}{\eta}$,

$$\boxed{\psi = \sqrt{\nu u x} F(\eta)}$$

Finding derivative transformations:

$$\frac{\partial \eta}{\partial x} \Big|_y = y \sqrt{\frac{u}{\nu}} \left(-\frac{1}{2} x^{-3/2} \right) = -\frac{\eta}{2x}$$

$$u = \frac{\partial \psi}{\partial y} = \sqrt{\nu u x} F' \sqrt{\frac{u}{\nu x}} = u F' = \left(\frac{\partial \psi}{\partial x} \frac{\partial \eta}{\partial y} + \frac{\partial \psi}{\partial y} \frac{\partial \eta}{\partial y} \right)$$

$$v = -\frac{\partial \psi}{\partial x} = -\left[\sqrt{\nu u} \frac{1}{2x^{-1/2}} F + \sqrt{\nu u x} F' \left(-\frac{\eta}{2x} \right) \right] = -\frac{1}{2} \sqrt{\frac{\nu u}{x}} (F + \eta F')$$

$$\psi_{yy} = \frac{\partial u}{\partial y} = \frac{\partial}{\partial y} \left(\frac{\partial \psi}{\partial x} \frac{\partial \eta}{\partial y} + \frac{\partial \psi}{\partial \eta} \frac{\partial \eta}{\partial y} \right) = \left(\frac{\partial}{\partial x} \frac{\partial \psi}{\partial y} + \frac{\partial}{\partial \eta} \frac{\partial \psi}{\partial y} \right) \left(\frac{\partial \eta}{\partial y} \right) = \left(\frac{\partial \eta}{\partial y} \right)^2 \frac{\partial^2 \psi}{\partial \eta^2} = \frac{u}{\nu x} F''(\eta)$$

$$\psi_{yy} = \sqrt{\frac{u}{\nu x}} u F''$$

$$\psi_{yyy} = u \frac{u}{2x} F'''$$

$$\psi_{yx} = u F' - \frac{\eta}{2x}$$

Insert into ψ eqn:

$$\psi \psi_{yx} - \psi_x \psi_{yy} = \nu \psi_{yyy}$$

$$- \frac{\eta u^2}{2x} F' F'' + \left[\frac{1}{2} \sqrt{\frac{u \eta}{\nu x}} (\eta F' - F) \right] \left[u \sqrt{\frac{u}{\nu x}} F'' \right] = \nu \frac{u^2}{\nu x} F'''$$

$$-\frac{1}{2} F F'' = F'''$$

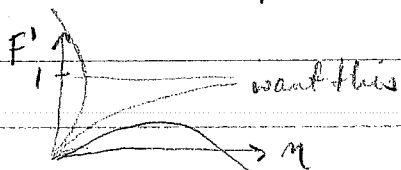
$$F''' + \frac{1}{2} F F'' = 0$$

BC:

$$y=0 \Rightarrow \eta=0 : F=0, F'=0$$

$$y \rightarrow \infty \Rightarrow \eta \rightarrow \infty : F' \rightarrow 1$$

The final analysis is done numerically using Runge-Kutta if $F''(0)$ is given. If $F''(0)$ is unknown, use shooting method.



Choose $F''(0)$ & march to ∞ & compare F' with 1.

$$\Rightarrow F''(0) = 0.33206$$

Rewriting η

$$\eta = y \sqrt{\frac{u_\infty}{\nu x^2}}$$

$$\eta = \frac{y}{x} Re_x^{1/2} \quad \text{where } Re_x = \frac{u_\infty x}{\nu}$$

If $o(y) \sim o(x)$ BL theory will fail!! ($\eta \approx 50$)

$\eta \rightarrow \infty$, $F' \rightarrow 1$ means $F \approx \eta + \epsilon$. Insert into pde,

$$\epsilon''' + \frac{1}{2}(\eta + \epsilon)\epsilon'' = 0$$

$$\epsilon''' + \frac{1}{2}\eta\epsilon'' \approx 0$$

$$(e^{\frac{\eta^2}{4}} \epsilon'')' \approx 0$$

$$A = e^{\int \frac{1}{2}\eta d\eta}, \quad (A\epsilon'')' = 0$$

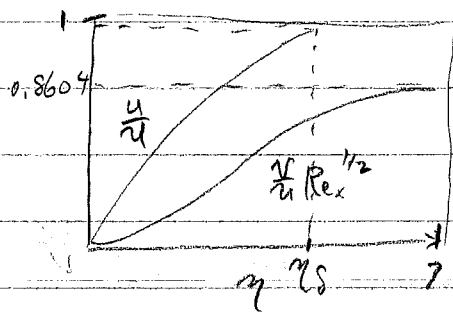
$\epsilon'' \sim c_1 e^{-\eta^2/4}$, the asymptotic value of ϵ .

$$\epsilon' \sim c_2 \eta e^{-\eta^2/4}$$

BL dies as $e^{-\eta^2/4}$

ex. if $\eta = 10$, $e^{-25} \Rightarrow$ BL approx falls apart!!

for Blasius, $\eta < 8$.



for Blasius, $\frac{u}{U} = 0.99$ at $\eta_\delta = 5$

$$\frac{\delta}{x} = 5 Re_x^{-1/2}$$

Shear Stress for laminar flow

$$\tau_w = \mu \left. \frac{\partial u}{\partial y} \right|_{y=0} = \mu \sqrt{\frac{u}{vx}} U F''(0)$$

$$\tau_w = \frac{\mu U}{x} Re_x^{-1/2} F''(0)$$

$$\tau_w = \frac{1}{2} \rho U^2 C_f, \quad C_f \text{ the friction coefficient}$$

$$\frac{C_f}{2} = Re_x^{-1/2} F''(0)$$

The drag (D) is

$$D = \int_0^x \tau_w dx$$

$$= \mu \sqrt{\frac{U}{\nu}} U F''(0) 2x^{1/2}$$

$$D \equiv \frac{1}{2} \rho U^2 \bar{C}_f x, \quad \bar{C}_f \text{ the average friction coefficient.}$$

$$\frac{\bar{C}_f}{2} = C_f = 0.664 Re_x^{-\frac{1}{2}}$$

Displacement Thickness (δ_1)

$$U \delta_1 = \int_0^{\infty} u dy \quad \xleftarrow{\text{fluid displaced}} \quad \int_0^{\infty} u dy \quad \xleftarrow{\text{inviscid}} \quad \int_0^{\infty} u dy \quad \xleftarrow{\text{viscous}}$$

$$\delta_1 = \int_0^{\infty} \left(1 - \frac{u}{U}\right) dy$$

$$\delta_1 = \sqrt{\frac{\nu x}{U}} \int_0^{\infty} (1 - F') d\eta$$

$$\delta_1 = \sqrt{\frac{\nu x}{U}} \lim_{\eta \rightarrow \infty} (\eta - F)$$

← find numerically.

$$\delta_1 = 1.7208 \sqrt{\frac{\nu x}{U}}$$

$$\frac{\delta_1}{x} = 1.72 Re_x^{-\frac{1}{2}}$$

The momentum thickness δ_2

$$U^2 \delta_2 \equiv \int_0^{\infty} u(U - u) dy$$

$$\delta_2 = \sqrt{\frac{\nu x}{U}} \int_0^{\infty} F'(1 - F') d\eta$$

Look at integral parts

$$\int F'(1-F') = F(1-F') + \int F(-F'') dy$$

$$= F(1-F') - 2F''|_0^\infty$$

$$\delta_2 = 2F''(0) \sqrt{\frac{\nu x}{u}}$$

$$\frac{\delta_2}{x} = 0.664 Re_x^{-\frac{1}{2}}$$

Energy Equation - Temperature

$$\theta = \frac{T - T_w}{T_\infty - T_w} \quad \frac{T_w}{T_\infty} \quad Pr = \frac{\nu}{\alpha}$$

poles

$$u \frac{\partial \theta}{\partial x} + v \frac{\partial \theta}{\partial y} = \frac{\nu}{Pr} \frac{\partial^2 \theta}{\partial y^2}$$

BC

$$y=0: \theta=0 \quad y \rightarrow \infty: \theta=1$$

If $Pr=1$, $\theta = \frac{u}{u_\infty}$ Reynolds Analogy

Solution is same as velocity solution with same pole's!!

$$q_w = -k \frac{\partial T}{\partial y}|_{y=0}$$

$$\tau_w = \mu \frac{\partial u}{\partial y}|_{y=0}$$

$$\frac{q_w}{k(T_\infty - T_w)} = \frac{\partial \theta}{\partial y}|_{y=0}$$

$$\frac{\tau_w}{\mu u_\infty} = \nu \frac{\partial (\frac{u}{u_\infty})}{\partial y}|_{y=0}$$

for $Pr = 1$,

$$\frac{\tau_w}{\mu U} = \frac{q_w}{k(T_w - T_\infty)}$$

$$\tau_w = \frac{1}{2} \rho U^2 C_f$$

$$q_w = h(T_w - T_\infty)$$

We get

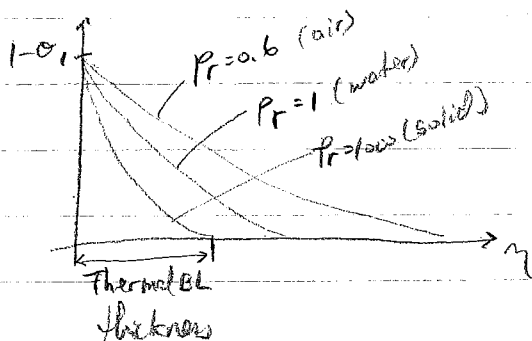
$$\frac{C_f}{2} = \frac{Nu}{Re}, \quad Nu = \frac{hL}{k} \text{ \& } Re = \frac{UL}{\nu}$$

If $Pr \neq 1$, we take $\theta(\eta)$,

$$\theta'' + \frac{Pr}{2} F \theta' = 0$$

$$\theta(0) = 0, \quad \theta(\infty) = 1$$

[If $\theta = F'$, & $Pr = 1 \Rightarrow$ we get Blasius Egn]



$\theta'(0)$ is the heat flux!

Solving, we find

$$\frac{S_T}{S} \approx Pr^{-1/3}, \quad 0.6 \leq Pr \leq 10$$

where S_T is defined at $1-\theta=0.01$

$$\theta'(0) = \begin{cases} 0.564 Pr^{1/2}, & Pr < 0.05 \\ 0.332 Pr^{1/3}, & 0.6 \leq Pr \leq 10 \\ 0.339 Pr^{1/3}, & Pr > 10 \end{cases}$$

$$q_w = -k \frac{\partial T}{\partial y} \Big|_{y=0} = h(T_w - T_\infty)$$

$$h = k \sqrt{\frac{u}{\nu x}} \theta'(0)$$

$$Nu_x = Re_x^{1/2} \theta'(0)$$

Viscous Dissipation

Important for high speed or highly viscous flow.

Look at flat plate case only.

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$$

$$u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} = \nu \frac{\partial^2 u}{\partial y^2}$$

viscous dissipation

$$u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y} = \alpha \frac{\partial^2 T}{\partial y^2} + \frac{\nu}{c_p} \left(\frac{\partial u}{\partial y} \right)^2$$

BC

for $y=0$, $u=v=0$

either $(T=T_w)$ or $(\frac{\partial T}{\partial y}=0 \text{ \& } T_w=?)$

$$y \rightarrow \infty, u \rightarrow U$$

$$T \rightarrow T_\infty$$

Similarity transformation is

$$\eta = y \sqrt{\frac{U}{\nu x}}$$

$$\psi = \sqrt{\nu x U} F(\eta)$$

From last time, we get the Blasius system

$$\boxed{\begin{aligned} F''' + \frac{1}{2} F F'' &= 0 \\ F(0) = F'(0) &= 0 \\ F'(\infty) &= 1 \end{aligned}} \quad \text{need } F''(0) = ?$$

$$u = \psi_y = \nu F'(\eta)$$

$$v = -\psi_x = \frac{1}{2} \sqrt{\frac{\nu x}{U}} (\eta F' - F)$$

Now, we look at temperature.
Can we relate the equation

$$T(x, y) \stackrel{?}{=} T(\eta)$$

$$\eta_x = y \sqrt{\frac{U}{\nu}} \cdot -\frac{1}{2} x^{-3/2} = -\frac{\eta}{2x}$$

Inserting terms into energy equation:

$$\begin{aligned} \nu F' \frac{dT}{d\eta} \left(-\frac{\eta}{2x} \right) + \frac{1}{2} \sqrt{\frac{\nu x}{U}} (\eta F' - F) \cdot \frac{dT}{d\eta} \sqrt{\frac{U}{\nu x}} &= \\ &= \alpha \frac{\nu}{\nu x} \cdot \frac{d^2 T}{d\eta^2} + \frac{\nu}{c_p} \left\{ \nu F'' \sqrt{\frac{U}{\nu x}} \right\}^2 \end{aligned}$$

Simplifying,

$$\frac{1}{2} \frac{u}{x} (-F) \frac{dT}{dy} = \frac{1}{Pr} \frac{u}{x} \frac{d^2 T}{dy^2} + \frac{u}{x c_p} F''^2$$

$$\frac{d^2 T}{dy^2} + \frac{Pr}{2} F \frac{dT}{dy} = -Pr \frac{u^2}{c_p} F''^2$$

The equation is linear, $\therefore$ we can use superposition.

$$\theta_1'' + \frac{Pr}{2} F \theta_1' = 0, \quad \theta_1(0) = 1 \quad \text{Need non-homog. BC} \\ \theta_1(\infty) = 0 \quad \text{with homogeneous eqn.}$$

$$\theta_2'' + \frac{Pr}{2} F \theta_2' = -2Pr F''^2, \quad \theta_2'(0) = 0 \\ \theta_2(\infty) = 0$$

The solution will be a linear combination of $\theta_1 + \theta_2$.

Let

$$g(y) = \frac{u^2}{2c_p} \theta_2$$

$$g'' + \frac{Pr}{2} F g' = \frac{u^2}{2c_p} \left[\theta_2'' + \frac{Pr}{2} F \theta_2' \right] \\ = -\frac{u^2}{c_p} Pr F''^2$$

So we use solution

$$T - T_\infty = C \theta_1 + \frac{u^2}{2c_p} \theta_2$$

where C is dependent on wall conditions.

$$\Theta_2(0) = f(Pr) \text{ only } \approx \begin{cases} Pr^{1/2} & 0.5 \leq Pr \leq 47 \\ 1.9 Pr^{1/3} & Pr > 47 \end{cases}$$

A) At $\frac{\partial T}{\partial y}|_{y=0} = 0$, 1st wall condition

$$T = \frac{u^2}{2c_p} \Theta_2(\eta)$$

$$T_{aw} - T_\infty = \frac{u^2}{2c_p} \Theta_2(0)$$

$$\frac{T_{aw} - T_\infty}{u^2/2c_p} = \Theta_2(0) = r, \text{ the recovery factor.}$$

B) $T|_{y=0} = T_w$, 2nd wall condition

$$T - T_w = C\Theta_1 + \frac{u^2}{2c_p} \Theta_2$$

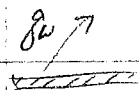
$$T_w - T_\infty = C(1) + (T_{aw} - T_\infty) \text{ where } \Theta_1(0) = 1, \text{ a BC for } \Theta_1.$$

$$T - T_\infty = (T_w - T_{aw})\Theta_1 + \frac{u^2}{2c_p} \Theta_2$$

$$C = T_w - T_{aw}$$

The heat flux out q_w from plate to fluid is

$$q_w = -k \frac{\partial T}{\partial y}|_{y=0}$$



$$q_w = -k \frac{\partial T}{\partial y} \Big|_{y=0} \sqrt{\frac{u}{\nu x}}$$

$$q_w = -k \sqrt{\frac{u}{\nu x}} (T_w - T_\infty) \frac{d\theta_1}{dy} \Big|_{y=0}$$

If $T_w > T_\infty$, heat transfer from plate into fluid.

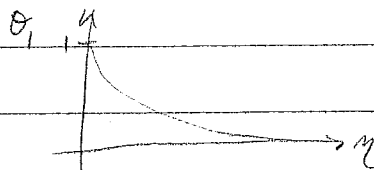
If $T_w < T_\infty$, " " " fluid into plate.

i.e. $(T_w - T_\infty)$ drives direction of H.T.

We can also get Nusselt Number

$$q_w = -k \sqrt{\frac{u}{\nu x}} (T_w - T_\infty) \frac{d\theta_1}{dy} \Big|_{y=0} \equiv h (T_w - T_\infty)$$

$$Nu_x = \frac{hx}{k} = - \frac{d\theta_1}{dy} \Big|_{y=0} Re_x^{1/2}$$



Solving $\theta_1 \neq \theta_2$

Solvable by many methods !!

Before solving $\theta_1 \neq \theta_2$, we need to find F .
So, first solve

$$f''' + \frac{1}{2} f f'' = 0.$$

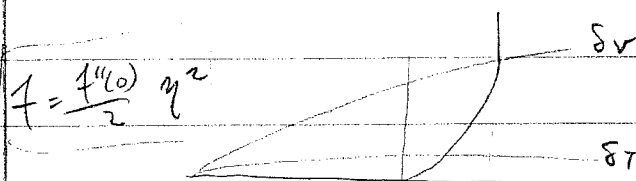
$$f(0) = f'(0) = 0$$

$$f'(\infty) = 1$$

$$f''(0) = ?$$

by shooting method. The f system is nonlinear!

Look at $Pr \rightarrow \infty$.



Because δ_T is very thin, $\frac{u}{u_\infty}$ will just be seen as linear $\Rightarrow f \propto \eta^2$

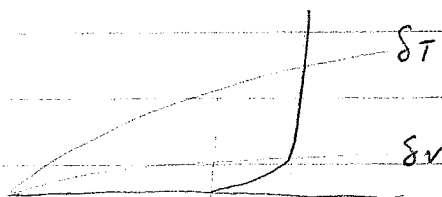
$$\frac{u}{u_\infty} = f'(\eta)$$

$$= f'(0) + f''(0)\eta + O(\eta^2)$$

Taylor series.

and $Pr \rightarrow 0$ (liquid metals)

HW problem



$$\frac{u}{u_\infty} = 1 \Rightarrow f \propto \eta$$

$$f = \eta, \text{ "plug flow"}$$

Similarity Transformation into the Falkner-Skan equation

$$\eta = y \sqrt{\frac{U}{\nu x}} \quad \psi = \sqrt{\nu U x} F(\eta) \quad , \quad U = C x^m$$

We wish to transform

$$U \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} = -\frac{1}{\rho} \frac{\partial p}{\partial x} + \nu \frac{\partial^2 u}{\partial y^2}$$

from Potential flow Theory, $-\frac{1}{\rho} \frac{\partial p}{\partial x} = U \frac{\partial U}{\partial x}$,

$$U \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} = U \frac{\partial U}{\partial x} + \nu \frac{\partial^2 u}{\partial y^2} \quad , \quad \begin{aligned} u(y=0) &= v(y=0) = 0 \\ u(y \rightarrow \infty) &= U \end{aligned}$$

Transforming each term:

$$\begin{aligned} U \frac{\partial u}{\partial y} &= \frac{\partial \psi}{\partial x} \frac{\partial x}{\partial \eta} + \frac{\partial \psi}{\partial \eta} \frac{\partial \eta}{\partial y} \\ &= \cancel{\frac{\partial \psi}{\partial x}} \left[\sqrt{\nu U x} F' \right] \left[\sqrt{\frac{U}{\nu x}} \right] \end{aligned}$$

$$\star \quad u = U F'$$

$$\frac{\partial u}{\partial x} = \frac{\partial}{\partial x} \left(\frac{\partial \psi}{\partial y} \right) = \frac{\partial}{\partial x} (U F') = \frac{\partial U}{\partial x} F' + \frac{\partial F'}{\partial x} U$$

$$\text{where } \frac{\partial U}{\partial x} = C m x^{m-1} = \cancel{\frac{m}{x} C x^m} = \frac{m}{x} U$$

$$\frac{\partial F'}{\partial x} = \frac{\partial F'}{\partial \eta} \frac{\partial \eta}{\partial x} = F'' \frac{\partial \eta}{\partial x} \quad \cancel{\frac{\partial F'}{\partial x} U}$$

$$\text{since } \cancel{\frac{\partial \eta}{\partial x}} \eta = y \sqrt{\frac{U}{\nu x}} = y \sqrt{\frac{C x^m}{\nu x}}$$

Dividing through into the following form

$$m x^2 = 15 \quad \eta = y \sqrt{\frac{c}{y}} x^{m-1} \quad \frac{F}{x^2} = 1$$

$$\eta = y \sqrt{\frac{c}{y}} x^{\frac{m-1}{2}}$$

$$\frac{\partial \eta}{\partial x} = y \sqrt{\frac{c}{y}} \frac{m-1}{2} x^{\frac{m-3}{2}} \quad \frac{\partial \eta}{\partial x} = 0$$

$$\frac{\partial \eta}{\partial x} = y \sqrt{\frac{c}{y}} \frac{m-1}{2} x^{\frac{m-3}{2}}$$

$$\frac{\partial \eta}{\partial x} = \frac{m-1}{2} y \sqrt{\frac{c}{y}} x^{\frac{m-3}{2}}$$

$$\frac{\partial \eta}{\partial x} = \frac{m-1}{2} \frac{\eta}{x}$$

$$\frac{\partial F}{\partial x} = \frac{\partial F}{\partial \eta} \frac{\partial \eta}{\partial x} = F' \frac{m-1}{2} \frac{\eta}{x}$$

$$\frac{\partial \eta}{\partial x} = \frac{\partial \eta}{\partial x} F' + \frac{\partial F}{\partial x} \eta$$

$$= \frac{m}{x} \eta F' + \frac{m-1}{2} \frac{\eta}{x} F'' \eta$$

$$\star \frac{\partial \eta}{\partial x} = \frac{m}{x} \eta F' + \frac{m-1}{2} \eta \frac{\eta}{x} F''$$

$$\text{OR } \frac{\partial \eta}{\partial x} = \frac{m}{x} \eta F' + \frac{m-1}{2} \eta \frac{\eta}{x} F''$$

$$\frac{\partial \eta}{\partial x} = \frac{\partial}{\partial x} \left(\frac{\partial \eta}{\partial y} \right) = \left(\frac{\partial}{\partial x} \eta + \frac{\partial}{\partial y} \left(\frac{\partial \eta}{\partial x} \right) \right) \left(\frac{\partial \eta}{\partial x} \frac{\partial \eta}{\partial y} + \frac{\partial \eta}{\partial y} \frac{\partial \eta}{\partial x} \right)$$

* Calculate each term separately! do not combine parenthesis until derivatives are calculated!!

$$-V = \frac{\partial \psi}{\partial x} = \frac{\partial \psi}{\partial x} \frac{\partial x'}{\partial x} + \frac{\partial \psi}{\partial \eta} \frac{\partial \eta}{\partial x}$$

(cancel gradient)

$$\text{since } \psi = \sqrt{VUx} F(\eta) \quad \& \quad U = Cx^m \quad \frac{V}{C} = V$$

$$\psi = \sqrt{VCx^{m+1}} F = F \sqrt{VC} x^{\frac{m+1}{2}}$$

$$\frac{\partial \psi}{\partial x} = F \sqrt{VC} \frac{m+1}{2} x^{\frac{m-1}{2}} \quad \frac{V}{C} = V$$

$$= F \frac{m+1}{2} \sqrt{\frac{VCx^m}{x}} \quad \frac{V}{C} = V$$

$$\left(\frac{\partial \psi}{\partial x} \right) F \frac{m+1}{2} \sqrt{\frac{VU}{x}} = \left(\frac{VC}{C} \right) \frac{V}{C} = \frac{VC}{C}$$

$$\text{since } \eta = y \sqrt{\frac{U}{V}} x^{-\frac{1}{2}} \quad \& \quad U = Cx^m \quad \frac{V}{C} = V$$

$$\eta = y \sqrt{\frac{C}{V}} x^{\frac{m-1}{2}} = y \sqrt{\frac{C}{V}} x^{\frac{m-1}{2}} \quad \frac{V}{C} = V$$

$$\frac{\partial \eta}{\partial x} = \frac{m-1}{2} y \sqrt{\frac{C}{V}} x^{\frac{m-3}{2}} = \frac{y}{x} \sqrt{\frac{Cx^m}{V}} \frac{m-1}{2} \frac{V}{C}$$

$$\frac{\partial \eta}{\partial x} = \frac{m-1}{2} y \sqrt{\frac{C}{V}} x^{\frac{m-3}{2}} = \frac{y}{x} \sqrt{\frac{Cx^m}{V}} \frac{m-1}{2} \frac{V}{C}$$

$$\frac{\partial \eta}{\partial x} = \frac{m-1}{2} \frac{y}{x} \quad \frac{y}{x} = \frac{V}{C}$$

$$\frac{\partial \psi}{\partial \eta} = \sqrt{VUx} F'$$

Combining terms,

$$-v = \frac{\partial y}{\partial x} = \frac{m+1}{2} F \sqrt{\frac{v}{x}} + \left[\sqrt{v/x} F' \right] \left[\frac{m-1}{2} \frac{y}{x} \right]$$

$$= \sqrt{\frac{v}{x}} \frac{m+1}{2} \left[-F + \frac{m-1}{m+1} y F' \right]$$

$$-v = u \sqrt{\frac{v}{u x}} \frac{m+1}{2} \left[F - \frac{1-m}{1+m} y F' \right]$$

$$\star \quad v = u \sqrt{\frac{v}{u x}} \frac{m+1}{2} \left[-F + \frac{1-m}{1+m} y F' \right]$$

$$\frac{\partial y}{\partial y} = \frac{\partial}{\partial y} \left(\frac{\partial y}{\partial y} \right) = \left(\frac{\partial}{\partial x} \frac{\partial y}{\partial y} + \frac{\partial}{\partial y} \frac{\partial y}{\partial y} \right) (u F')$$

$$\star \quad \frac{\partial y}{\partial y} = u \sqrt{\frac{u}{v x}} F''$$

$$\star \quad \frac{\partial y}{\partial x} = \frac{\partial}{\partial x} (C x^m) = C m x^{m-1} = \frac{m}{x} C x^m = \frac{m y}{x}$$

$$\frac{\partial^2 y}{\partial y^2} = \frac{\partial}{\partial y} \left(\frac{\partial y}{\partial y} \right) = \frac{\partial}{\partial y} \left[u \sqrt{\frac{u}{v x}} F'' \right]$$

$$= \left(\frac{\partial}{\partial x} \frac{\partial y}{\partial y} + \frac{\partial}{\partial y} \frac{\partial y}{\partial y} \right) \left[u \sqrt{\frac{u}{v x}} F'' \right]$$

$$\star \quad \frac{\partial^2 y}{\partial y^2} = \frac{u^2}{v x} F'''$$

$$F' \sqrt{\frac{u}{v x}} = \frac{y}{x} F'$$

The boundary ^{layer} equation again is

$$u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} = \nu \frac{\partial^2 u}{\partial x^2} + \nu \frac{\partial^2 u}{\partial y^2}$$

Inserting transformed variables:

$$u F' \left[\frac{m}{x} u F' + \frac{m-1}{2} u \frac{\eta}{x} F'' \right] + \left[u \sqrt{\frac{\nu}{u x}} \frac{m+1}{2} \left(-F + \frac{1-m}{1+m} \eta F' \right) \right] \left[u \sqrt{\frac{\eta}{\nu x}} F'' \right] =$$

$$= u \left[\frac{2 \eta m}{x} \right] + \nu \left[\frac{2 \eta^2}{\nu x} F''' \right]$$

$$\frac{m}{x} F'^2 + \frac{m-1}{2} \frac{\eta}{x} F' F'' + \frac{m+1}{2x} F'' \left(-F + \frac{1-m}{1+m} \eta F' \right) = \frac{m}{x} + \frac{F'''}{x}$$

$$m F'^2 + \cancel{\frac{m-1}{2} \eta F' F''} + \cancel{\frac{1-m}{2} \eta F' F''} - \frac{m+1}{2} F F'' = m + F'''$$

$$F''' + m - m F'^2 + \frac{m+1}{2} F F'' = 0.$$

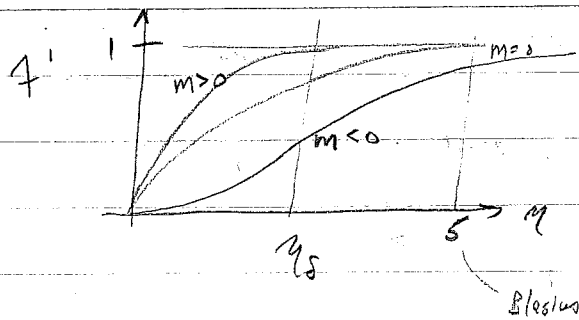
$$\boxed{F''' + \frac{m+1}{2} F F'' + m(1 - F'^2) = 0}$$

With BCs

$$F(0) = 0$$

$$F'(0) = 0$$

$$F'(\infty) = 1$$



adverse pressure gradient: $\frac{dP}{dx} > 0$, $m < 0$

favorable " " : $\frac{dP}{dx} < 0$, $m > 0$ thinner BL $\Rightarrow$ due to $\frac{\partial P}{\partial x}$.

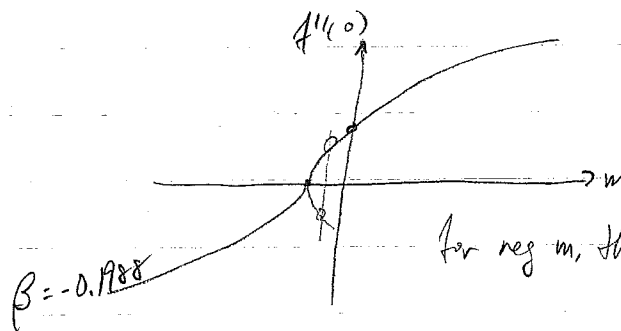
$$y|_{f'=0.99} = \eta_s \sqrt{\frac{c x^m}{\nu x}}$$

To solve Falkner-Skan equation, we need to first find $f''(0)$.

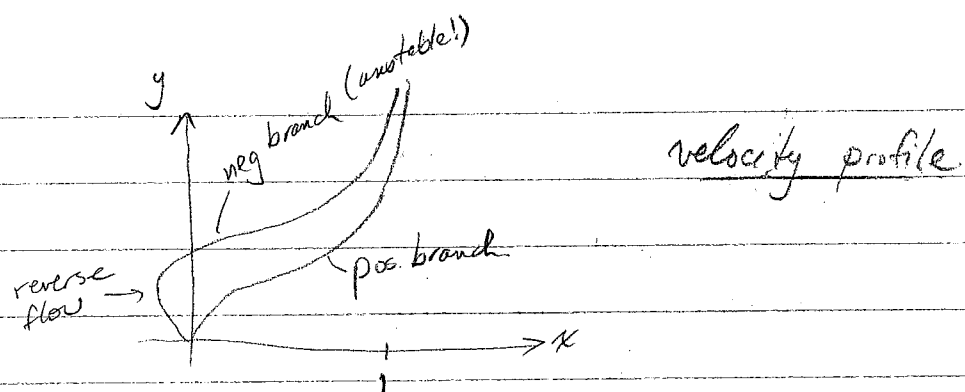
$$\tau_w = \mu \frac{\partial u}{\partial y} \Big|_{y=0} = \frac{\mu}{\rho \nu} \frac{u^2}{x} f''(0) \sqrt{\frac{u x}{\nu}}$$

$$\tau_w \equiv \frac{1}{2} c_f \rho u^2$$

$$c_{f/2} = f''(0) Re_x^{-1/2}$$



for neg m , there are two branches



For $m < 0$, there are many branches. Some of them are unstable.

Temperature

$$u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y} = \alpha \frac{\partial^2 T}{\partial y^2} + \frac{v}{c_p} \left(\frac{\partial u}{\partial y} \right)^2$$

$$T(x, 0) \stackrel{?}{=} T_w(x)$$

$$T(x, \infty) \stackrel{?}{=} T_\infty(x)$$

$$\theta = \frac{T - T_w}{T_\infty - T_w} \Rightarrow \theta(0) = 0, \quad \theta(\infty) = 1$$

But is $\theta(\eta)$?

Plugging θ into PDE for similarity transformations, we get 3 cases where $\theta(\eta)$.

case 1 $T_{\infty} - T_w = \text{const.}$

$$T_{\infty} = K_{\infty} + (T_{\infty} - T_w) \delta \ln x$$

$$T_w = K_w + (T_{\infty} - T_w) \delta \ln x$$

gives

$$\theta'' + \frac{m+1}{2} Pr \theta' - \gamma Pr \theta = 0 \quad \text{without viscous dissip. } \frac{\nu}{c_p} \left(\frac{\partial u}{\partial y} \right)^2$$
$$\Phi = 0$$

If $\delta = 0$, T_w & T_{∞} is constant.

$$\Phi = 0$$

wedge effect.

If $m = 0$, flat plate flow (Blasius flow).

can allow for viscous dissipation.

for $m = 0$ case, can do $\delta = 0$, with or without Φ

for $m \neq 0$ case, no Φ

for $m = 0$ & $\gamma \neq 0$, Temp. distribution T_{∞} & T_w
will be logarithmic.

case 2 $T_\infty = \text{const.}$
 $T_w = T_\infty + K_2 x^\delta$

$$\theta = \frac{T - T_\infty}{T_w - T_\infty}$$

$$\theta'' + \frac{m+1}{2} Pr \theta' - \delta Pr \theta = -Pr E f''^2 x^{2m-\delta}, \quad \theta(0) = 0, \quad \theta(\infty) = 1$$

when $E = \frac{c^2}{c_p K_2}$ where c is in $u = cx^m$

Most people ignore E term.

case 3 (most important)

$$T_\infty = \text{const.}$$

$$T_w = T_\infty - K_3 x^\delta$$

$$\theta = \frac{T - T_\infty}{T_\infty - T_w}$$

$$\theta'' + \frac{m+1}{2} Pr \theta' - \delta Pr \theta = -Pr E f''^2 x^{2m-\delta} \quad \theta(0) = 0, \quad \theta(\infty) = 1, \quad \theta'(0) = ?$$

Find heat flux.

$$q_w = -k \frac{\partial T}{\partial y} \bigg|_{y=0}$$

$$T = (T_\infty - T_w) \theta + T_\infty$$

$$q_w = -k (T_\infty - T_w) \theta'(0) \sqrt{\frac{c x^m}{\nu x}}$$

$$\delta_1 = \int_0^{\infty} \left(1 - \frac{v}{u}\right) dy$$

$$\delta_2 = \int_0^{\infty} \frac{v}{u} \left(1 - \frac{v}{u}\right) dy$$

$$\delta_3 = \int_0^{\infty} \frac{v}{u} \left(1 - \frac{T - T_w}{T_w - T_{\infty}}\right) dy$$

$$\delta_4 = \int_0^{\infty} \frac{v}{u} \left(1 - \frac{c - c_s}{c_s - c_{\infty}}\right) dy$$

Laminar BL's